

PAPER 1 / QUESTION 42

2222-6513 / original question 8

- 42.** Two blocks of different masses are released from identical springs of elastic constant $k = 100 \text{ Nm}^{-1}$, initially compressed a distance $\Delta x = 0.1 \text{ m}$. Block X has a mass of 1 kg and block Y has a mass of 0.25 kg .

What are the velocities of the blocks when they leave the springs?

	Velocity of block X	Velocity of block Y
A.	1.0 ms^{-1}	1.0 ms^{-1}
B.	2.0 ms^{-1}	1.0 ms^{-1}
C.	1.0 ms^{-1}	2.0 ms^{-1}
D.	2.0 ms^{-1}	2.0 ms^{-1}